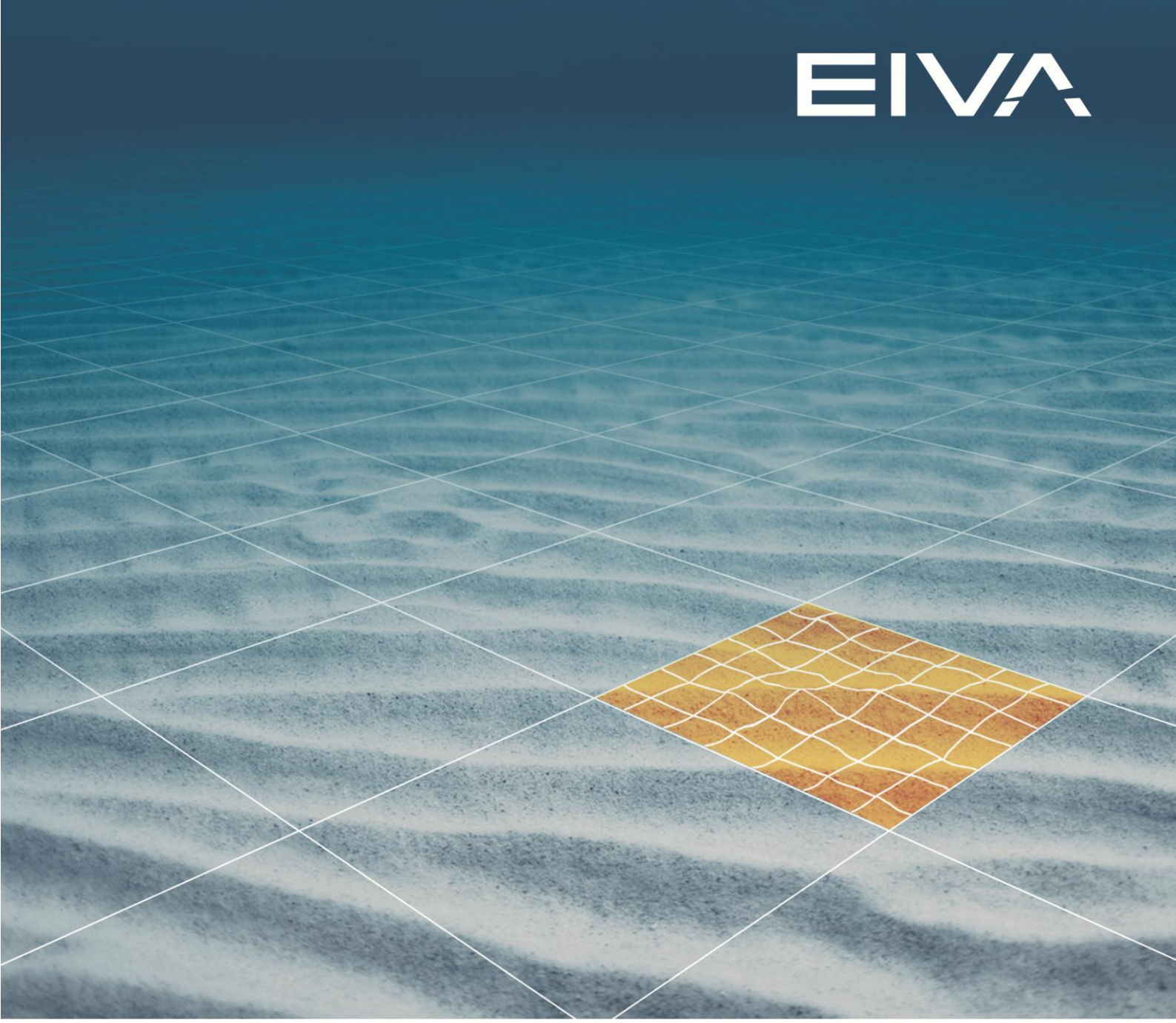

NAVISUITE KUDA CORE 4.8

QUICK START GUIDE

Last update: 24/11/2025
Version: 4.8

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1 Introduction

1.1 NaviSuite Kuda overview

NaviSuite Kuda is EIVA's dedicated software solution for small-scale, single-vehicle hydrographic operations using both crewed vehicles and unmanned surface vessels (USVs). It supports a wide range of sensors and introduces advanced functionalities for remote operation, including autopilot control.

NaviSuite Kuda combines components from the wider NaviSuite product line into a streamlined bundle:

- **NaviScan** – for data acquisition and configuration (offline and online)
- **Helmsman** – for user interface management

This integration ensures a focused system tailored for efficient setup, operation, and data collection. Moreover, it introduces many features to the user experience, such as workflow and UI.

There are four options available for NaviSuite Kuda

NaviSuite Kuda Core
NaviSuite Kuda Core + Processing
NaviSuite Kuda Pro
NaviSuite Kuda Pro + Processing

The two options of the product, NaviSuite Kuda Core and NaviSuite Pro, differ primarily on the intended use case. NaviSuite Kuda Core supports a single vehicle (eg a USV or a manned vessel), while NaviSuite Kuda Pro supports multiple vehicles (eg a manned vessel with one or more ROTVs).

While not covered in this document, there is also a data processing option available for each of these options,

NaviSuite Kuda requires a licence to one of these above-mentioned options of the software. You can read more about our licence types on our website.

This document will focus on NaviSuite Kuda Core.

1.2 Purpose of this guide

This quick start guide is designed to help users begin working with NaviSuite Kuda Core, without assuming prior experience with EIVA NaviSuite software. It highlights the features unique to NaviSuite Kuda Core and links to EIVA's knowledge base for in-depth documentation of the underlying NaviScan and Helmsman components.

Important: Do not install the NaviSuite Kuda Core bundle installer on a computer that already has a NaviPac or NaviScan install. Please uninstall NaviPac and NaviScan before installing Kuda Core. Likewise, once you have installed Kuda Core, do not install NaviPac or NaviScan as a standalone installation. These situations can lead to conflicts.

This document is separated into two main sections – the configuration of NaviSuite Kuda, ie setting up the project and vessel correctly, and the user interface and features accessed during NaviSuite Kuda operation.

For installation, activation and basic setup, see Getting Started with NaviSuite Kuda Core from EIVA's knowledge base.

Note: If you want NaviScan to be visible when launching NaviSuite Kuda Core Config, enable it under **Project Settings** > **Online** > **Show NaviScan**.

2 NaviSuite Kuda Core Config

NaviSuite Kuda Core Config is based on the NaviScan Config, but comes with a tailored UI and features. A more detailed explanation of how to use this configuration tool is available in the NaviScan Config Manual.

This process is where you will configure your vehicle, the instruments onboard, and the environment in which it will operate.

Below is a brief introduction to the main view of NaviSuite Kuda Core Config.

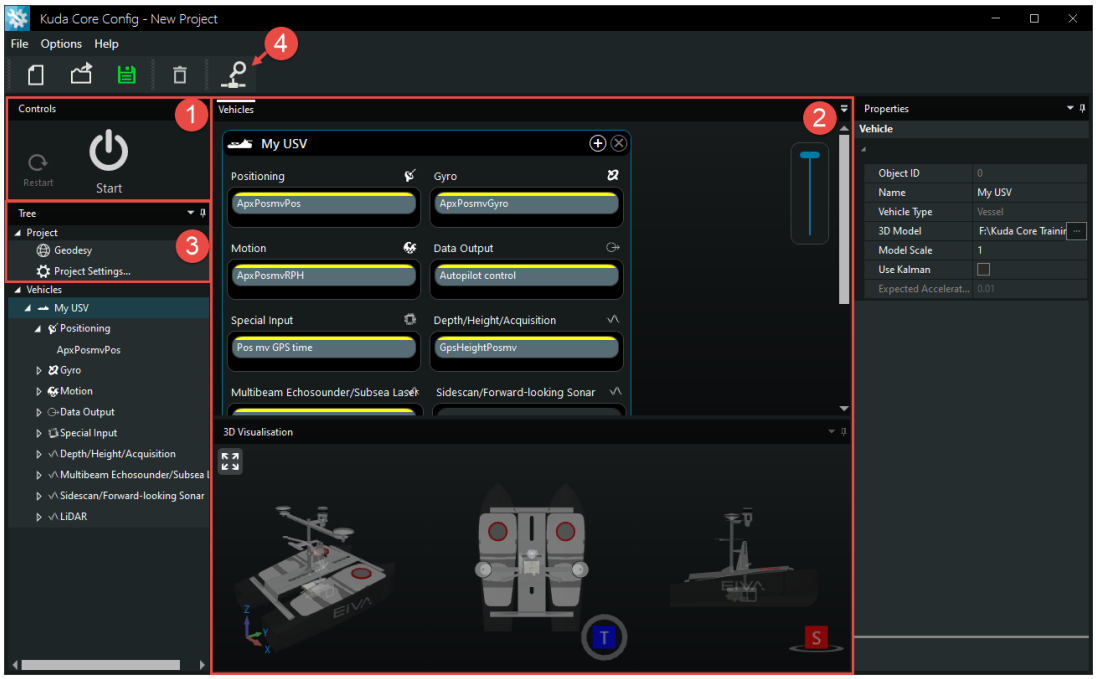


Figure 1 Highlighting the features of NaviSuite Kuda Core Config

1. **Controls** – Start, stop and restart NaviSuite Kuda Core Configuration.
2. **Vehicles** and **3D Visualiastion** – Lists the sensors at the top and shows a 3D model of the vehicle from multiple angles below.
3. **Tree** – Contains the Project and Vehicles in a nested tree structure:
    - **Project** – Geodesy and Project Settings.
    - **Vehicles** – A tree representation of the sensors and vehicle.
4. **Select instrument by port** – A tool to find instruments by the port they are assigned to.

2.1 The Vehicles section and 3D Vehicle Visualisation

The **Vehicles** section groups all existing sensors on a vehicle in a vehicle box. This helps for an easier visualisation of available sensors and drivers.

Note: NaviSuite Kuda Core supports only one vessel. Expanded vehicle support is a feature available in NaviSuite Kuda Pro.

With the vehicle selected from either the **Vehicles** tree or the **Vehicles** view, the properties of the vehicle are shown to the right under **Properties**. The properties include the **Name** of the vessel, which can be customised, as well as the option to choose a **3D model** and the **Model scale**. Additionally, you can select if you want to use calculate a Kalman-filtered position of the vehicle by toggling **Use Kalman**.

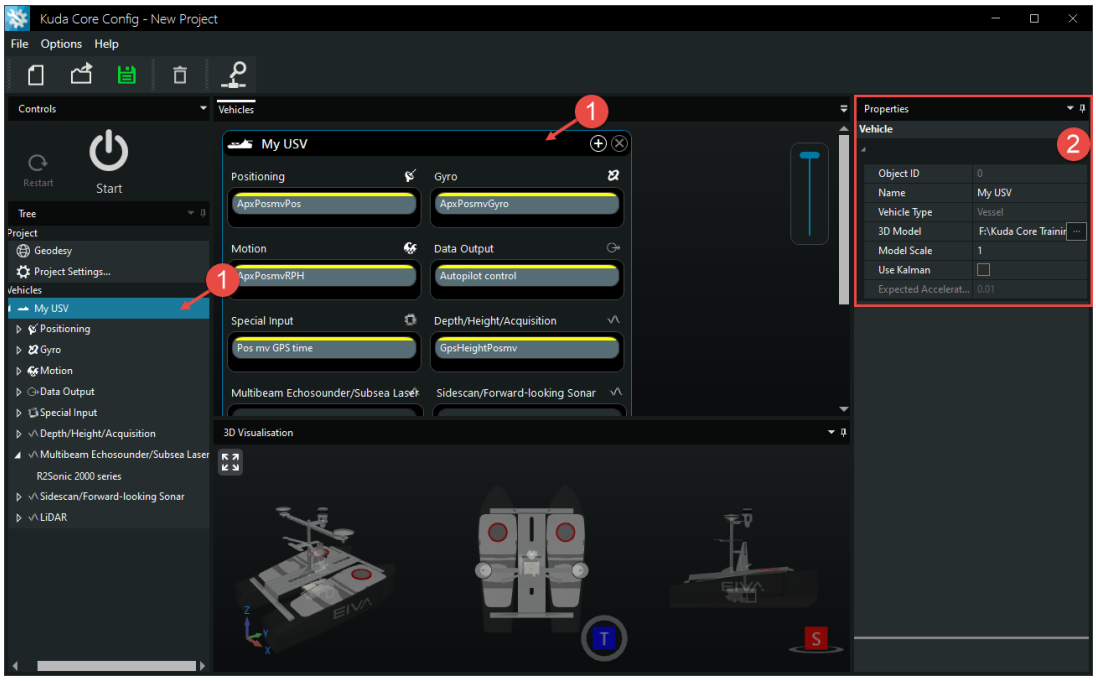


Figure 2 Accessing the vehicle properties (2) can be done via the **Vehicles** tree (1) or the Vehicles view (1)

2.2 Add a new instrument

In NaviSuite Kuda Core Config, you can add a sensor in several ways.

The first is by right-clicking on the vessel in the **Vehicles** tree and selecting **Add instrument.** You can also right-click within the **Vehicles** window.

Finally, you can also select the vehicle by left-clicking on the vehicle in the **Vehicles** window and clicking the **+** in the top right of the window.

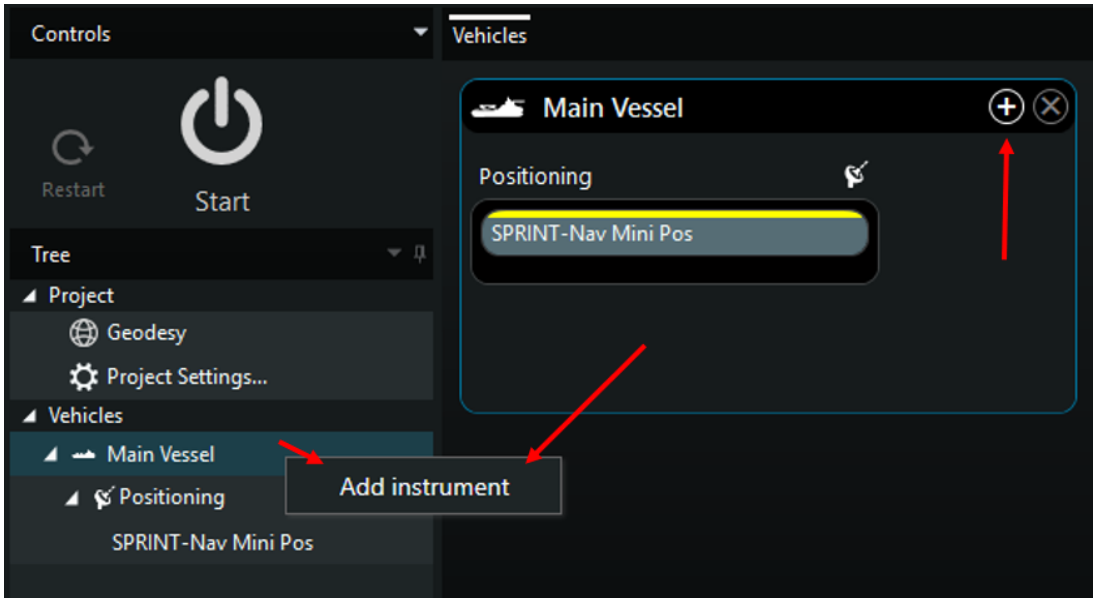


Figure 3 Add instrument by right-clicking or clicking the **+**

Either method will bring you to the **New Instrument** screen. Expand the **Instrument Categories** and select the proper instrument driver.

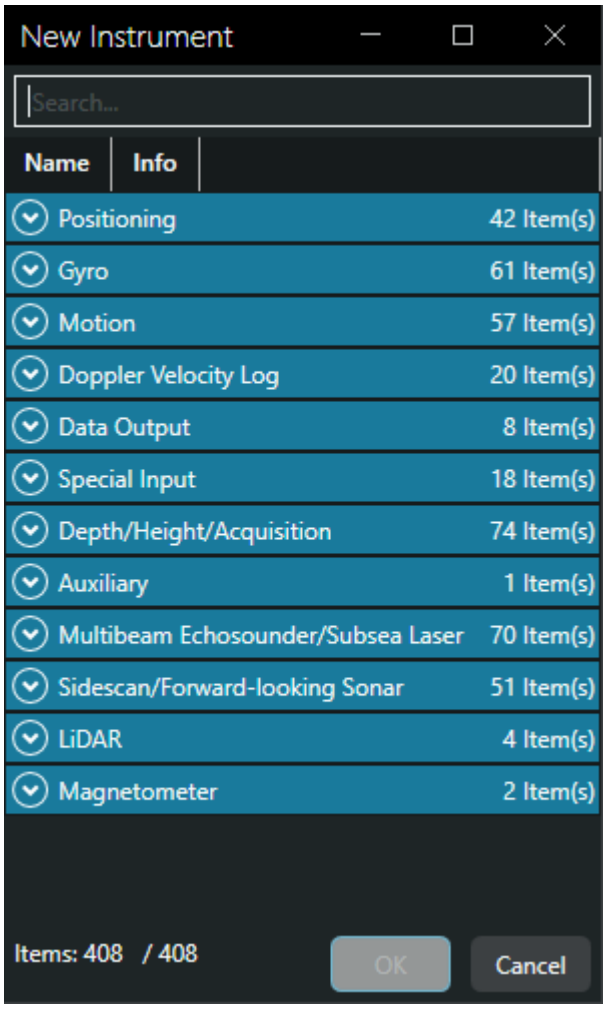


Figure 4 New Instrument window listing instrument drivers

2.3 Geodesy

NaviSuite Kuda Core Config creates a .GBIN geoid file, which improves handling of geoid files and handling of nodes without geoidal separation values in the matrix.

Refer to the NaviSuite Geodesy Manual for more information.

2.4 Project Settings

From **Project Settings**, you can manage systems parameters regarding positioning and data, online settings for ports and communication, extra information for configuring instruments and settings for the logs.

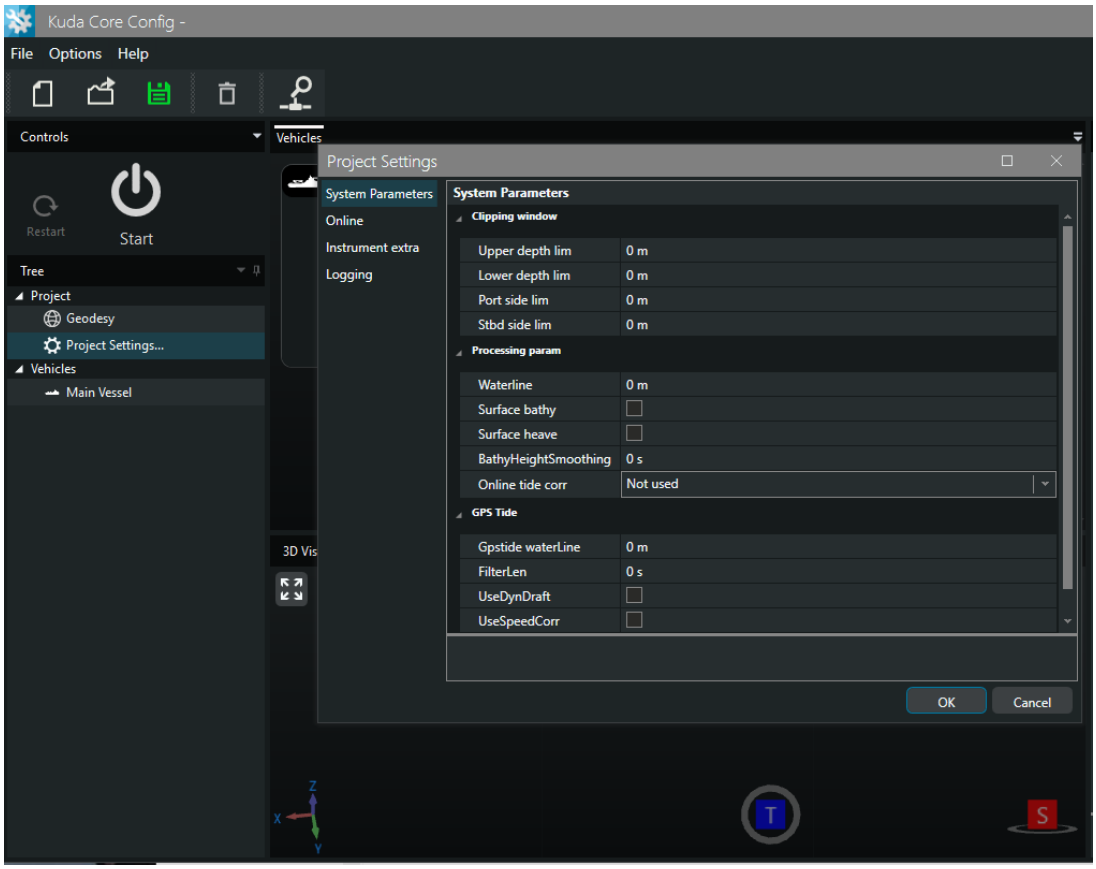


Figure 5 Project Settings window

2.4.1 System Parameters

For surface vehicles, we recommend enabling both **Surface bathy** and **Surface heave**, while for a subsea vehicle the typical setting is to use **Surface bathy** but not **Surface heave**.

Note: Within EIVA NaviSuite's concept of bathy, the legacy height component on subsurface vehicles was implemented using pressure sensors. Surface vehicles, before the modern RTK GNSS solutions, were positioned only from the water surface or tide information. With the generalisation of RTK GNSS heights, the height component of the navigation is now implemented as bathy drivers; inputs similar to pressure sensors but

coming from GPS devices (eg GGA GPSHeight). The category name has been changed to Depth/Height/Acquisition.

2.4.2 Online

In the **Online** section, under **Advanced,** you will find the option to **Show NaviScan**.

Selecting **Show NaviScan** will mean NaviScan Online will be started as a separate window together with the start of the acquisition, should it be necessary or desired.

By default, NaviScan runs in the background, with all tools necessary for survey being present in the main display – Helmsman.

2.4.3 Instrument Extra

The **Instrument Extra** tab contains the settings specific to individual sensor types that could apply to the project and not the driver for the instrument alone.

For example, the image below shows the settings to control an R2Sonic multibeam echo sounder from the NaviSuite Kuda Core **Sonar Controls** button:

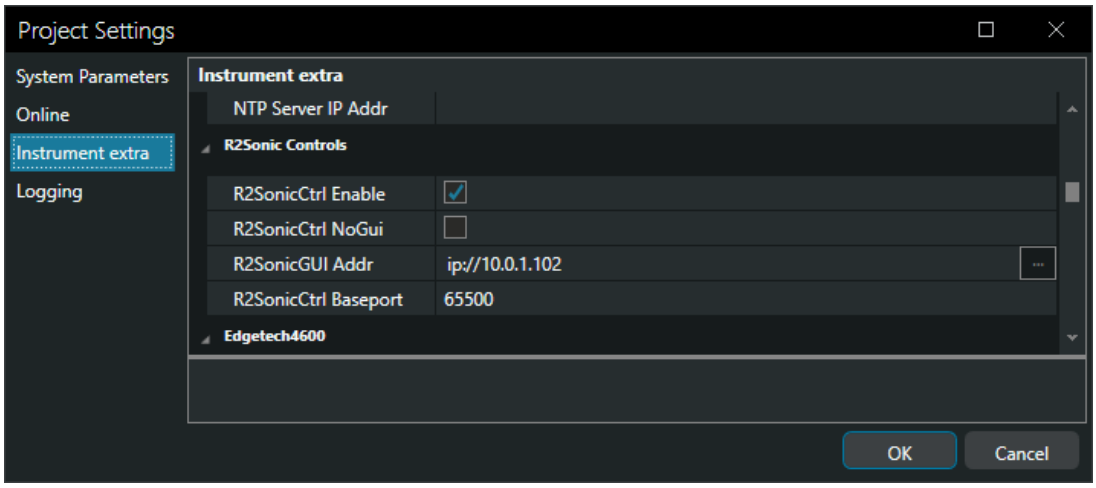


Figure 6 Example of Instrument Extra settings for controlling an R2Sonic multibeam echo sounder from the NaviSuite Kuda Core Sonar Controls

Sonar Control can also be opened from Helmsman, see section 3.3.

2.5 Select Instruments by Port

The **Select Instruments by Port** tool is an easy way to see which driver listens to which port in the NaviSuite Kuda Core Config.

Figure 7 Select Instruments by port button (1), the window showing instruments and ports (2)

3 NaviSuite Kuda Core main view

The NaviSuite Kuda Core main view is built on the NaviSuite UI, Helmsman. A manual detailing the user interface of Helmsman can be found on EIVA's knowledge base:

Helmsman 4 User Interface Manual

NaviSuite Kuda Core includes features and UI specifically designed for the expected use cases of the software.

Figure 8 A standard NaviSuite Kuda Core single-screen layout

3.1 Setup toolbar

The Setup toolbar contains two important buttons.

Start acquiring data with the **Start Sensor Acquisition** button in the NaviSuite Kuda Core main view.

Open settings for the sensors and paths by opening NaviSuite Kuda Core Config by pressing the **Sonar Setup** button.

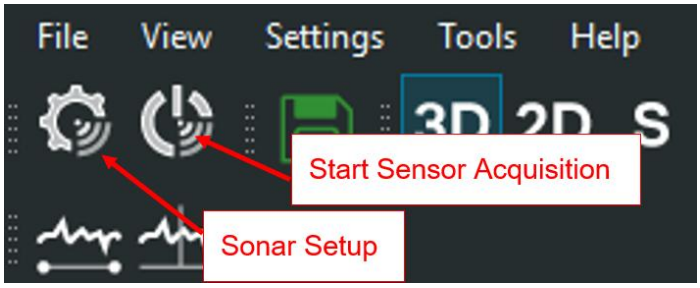


Figure 9 The Setup toolbar

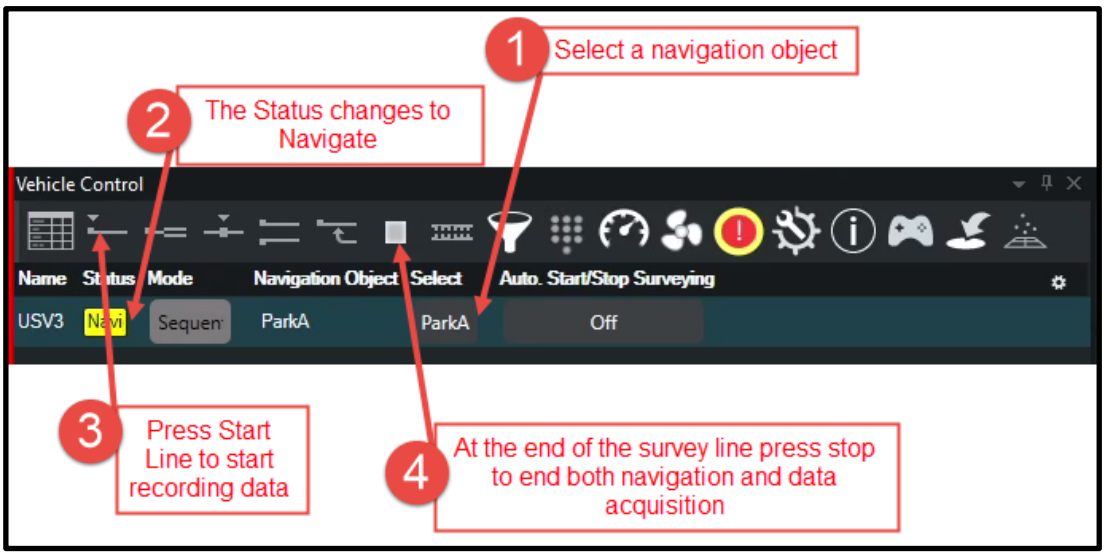
3.2 Vehicle Control

The **Vehicle Control** view is used to control the navigation of a vehicle that is interfaced with an autopilot as well as to record data files.

When a navigation object (eg waypoints and runlines) is selected, the status changes to *Navigate*.

When choosing a runline, both the colour under **Status** in the **Vehicle Control** view and the colour of the runline on the **Map View** change to yellow. In this situation, the vehicle navigates but does not yet record any data.

When pressing **Start Line** in the **Vehicle Control** view, the status changes to *Surveying* and the colour of the status section and the runline colour change to green.



The **Mode** column in the **Vehicle Control** refers to the handling of runline segments. There are three options:

Manual – the runline segments are changed manually by pressing on the **Next Segment** button
Automatic – the runline segment that is followed is chosen automatically as the closest segment of the runline to the vehicle
Sequential – the segment of the runline that is to be used is changed in consecutive order as the vehicle progresses along the runline.

Note: Working with *Sequential* is recommended. When using *Automatic*, if the vessel is told to perform a U-turn where the runline turn radius is less than the actual vessel turn radius, there is a risk that the start of the turn segment will be taken again when the vessel has finished the turn, thus making the vessel to go in circles.

3.2.1 Autopilot control overview

The **Vehicle Control** has additional buttons for controlling an autopilot:

1. **Set Speed**
2. **Start Thrusters**
3. **Emergency Stop**
4. **Set Auto Emergency Stop Time**
5. **Autopilot States**
6. **Joystick**
7. **Send Vehicle to**
8. **Build Online DTM**

Figure 10 Additional autopilot control buttons – the numbers in the figure corresponding to the numbers on the list

3.2.2 Set (Autopilot) Speed

The **Set Speed** button opens the **Set Autopilot Speed** window where you can set the desired speed of the vehicle as a percentage of the maximum speed of the thrusters.

Figure 11 Set Vessel Speed

3.2.3 Start Thrusters

The **Start Thrusters** button arms the vessel, therefore allowing for the autopilot to drive the vehicle.

3.2.4 Emergency Stop

The **Emergency Stop** button disarms the autopilot, stopping the thrusters and navigation.

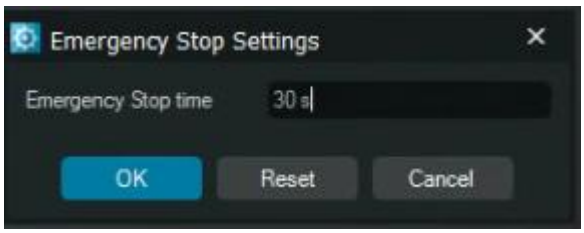


Figure 12 Set Emergency Stop time interval

3.2.5 Set Auto Emergency Stop Time

When operating under remote supervision, if the supervisor loses connection to the supervised onboard computer, the onboard supervised NaviSuite Kuda Core will continue to navigate. By setting the automatic emergency stop, this is prevented.

Clicking on the **Set Auto Emergency Stop Time** button in the **Vehicle Control** view will open a window, and you can set the time interval.

If the supervised computer feels it is unsupervised for a specific amount of time (**Emergency Stop Time**), it will disarm the autopilot, stopping the vessel until the supervisor re-arms the autopilot by hitting the **Start Thruster** button.

By default, the setting is *30 seconds* of lost communication between the supervisor and supervised computer. You should adjust the time interval depending on the network environment and situation.

3.2.6 Autopilot States

Autopilot States helps you to monitor the battery status of the onboard computer and autopilot operation modes.

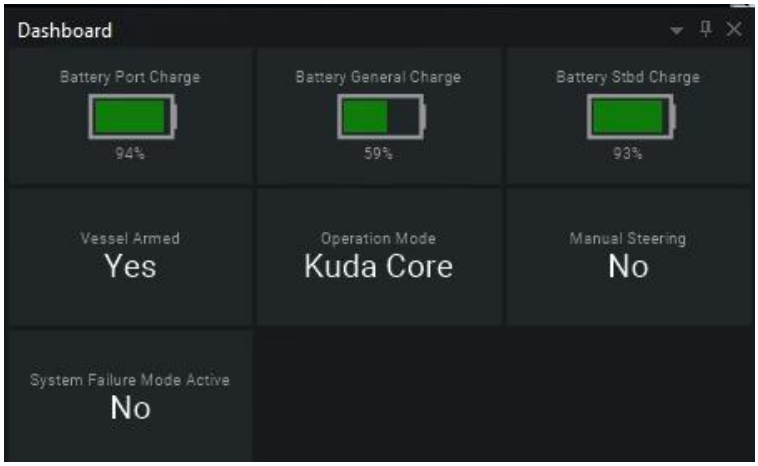


Figure 13 Autopilot States Dashboard

- **Vessel Armed** – Indicates whether the autopilot is armed or disarmed (Yes/No).
- **Operation Mode** – Specifies which control input the autopilot expects. Possible modes are:
  - **Idle** – The autopilot has just started, and neither the onsite operator (joystick) nor the NaviSuite Kuda Core operator has attempted control.
  - **Joystick** – The onsite operator controls the vessel using a joystick.

- **Manual (Manual with Heading)** –The autopilot maintains a constant heading while speed (forward/reverse) is manually adjusted. Set from either the joystick or NaviSuite Kuda Core.
- **Kuda Core** – The vessel is controlled directly from NaviSuite Kuda Core.

- **Manual Steering** – Shows whether the vessel is controlled manually or via software (Yes/No).
- **System Failure Mode Active** – Indicates whether a system fault has been detected (Yes/No).

3.2.7 Joystick

The **Joystick** option allows you to take manual control of the vehicle by setting up the desired heading and manually controlling forward or reverse thrust.

The actual heading of the vehicle is displayed as a gyro, and the desired heading can be dragged. The autopilot will then receive the command from NaviSuite Kuda Core and will adjust the vehicle heading to match the desired heading.

Below the heading is a slider where you can set the speed, either forward (to the right of centre) or reverse (to the left of centre).

There are two options: by default, the slider will keep the last setting, or if **Bounce to Zero** is ticked on then when you stop pressing on the mouse button to control the slider, it will automatically go back to zero thrust.

Hint: Set the **Map View** to north up so that the orientation of the vehicle in the **Map View** matches the one in the joystick for easier steering**.**

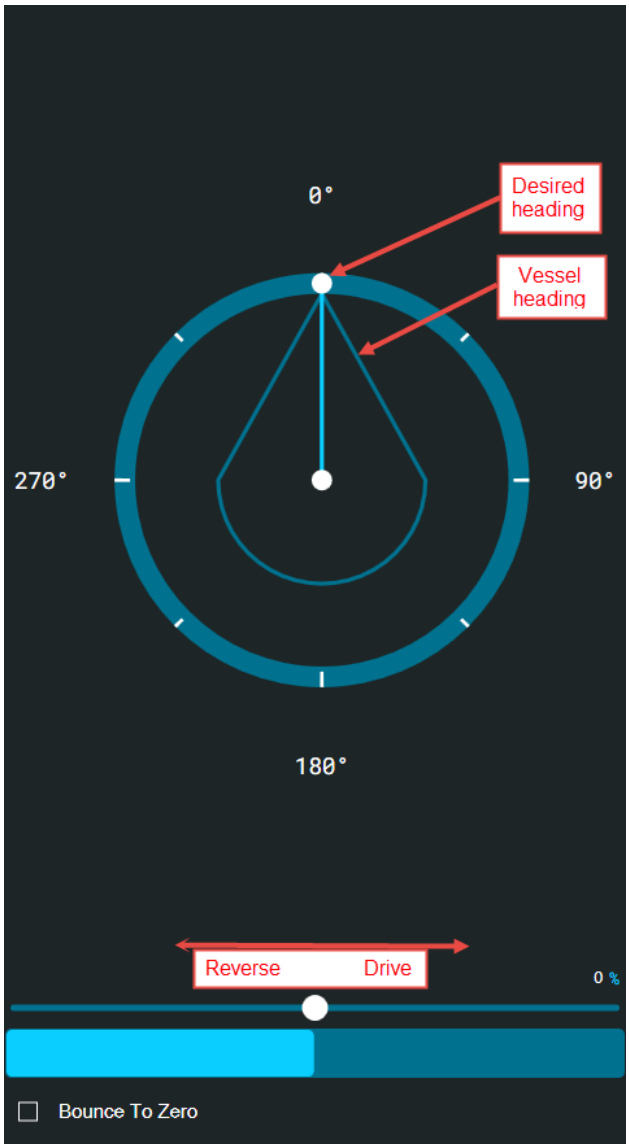


Figure 14 The manual joystick control

3.2.8 Send Vehicle To

The **Send Vehicle To** button is intended for fast vehicle control and is particularly useful if using the supervisor mode. By clicking on it, the mouse cursor becomes a digitising cursor and by clicking on the **Map View**, a waypoint is generated, automatically synchronised and sent from the supervisor to the supervised computer and set as navigation object so that the vessel will head towards it.

3.2.9 Build Online DTM

Build Online DTM replicates the feature to **Build an Online DTM** from the **Project Tree**.

Online DTM can be started if a user right clicks on **Project tree** > **Online** > **NaviScan**. Click on **Create Online** where user can choose between:

Online DTM
Point cloud
Online Cleaning

3.3 Sonar Controls toolbar

The **Sonar Controls** toolbar contains the options for sensors that are accessed during online operation (acquisition).

Once NaviSuite Kuda Core acquisition is started and the NaviScan kernel is connected to the project, the **Sonar Controls** toolbar appears. The sonar controls can be then opened, and the options can be adjusted.

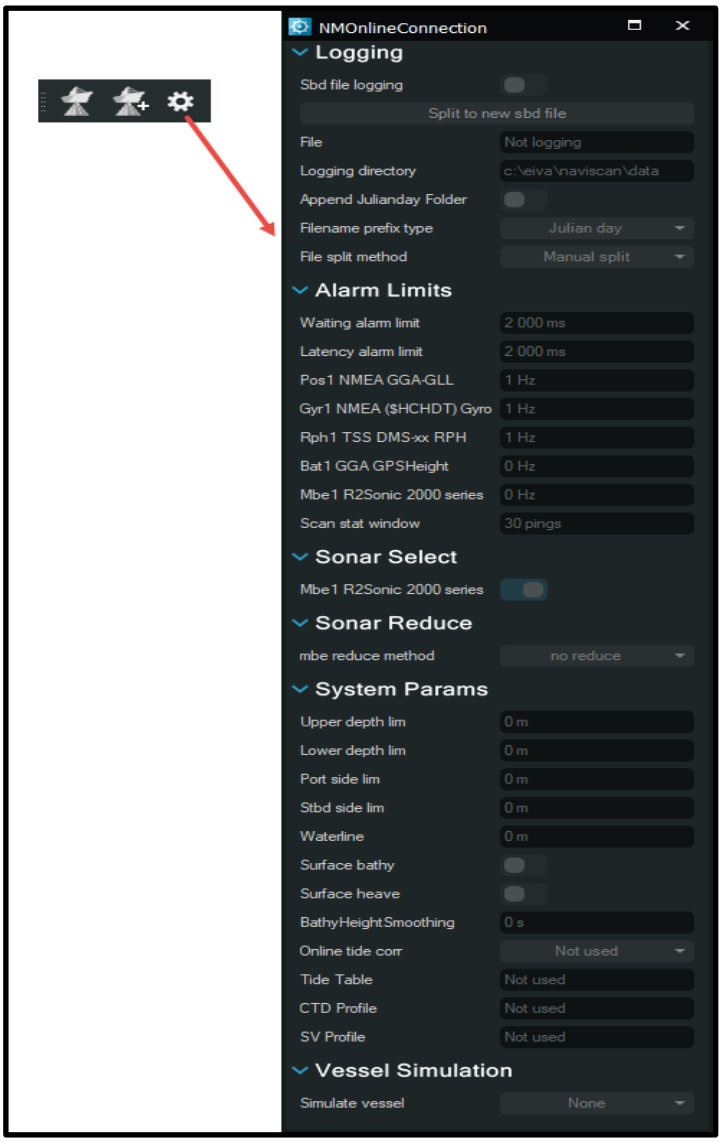


Figure 15 The NaviScan Controls window

If you want to control the multibeam from NaviSuite Kuda Core, set up the controls in the configuration on **Project Settings** > **Instrument Extra**, and you will have a section from where you can adjust the multibeam settings:

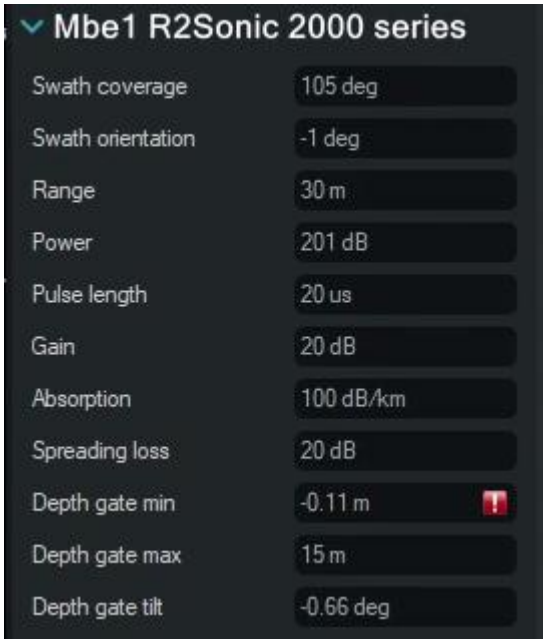


Figure 16 Example where an R2Sonic multibeam is controlled from NaviSuite Kuda Core

3.4 The NaviScan Profile

The **NaviScan Profile** view is made to display the real-time multibeam data as well as the multibeam wedge and gates. It can be added from **View** > **New NaviScan profile**.

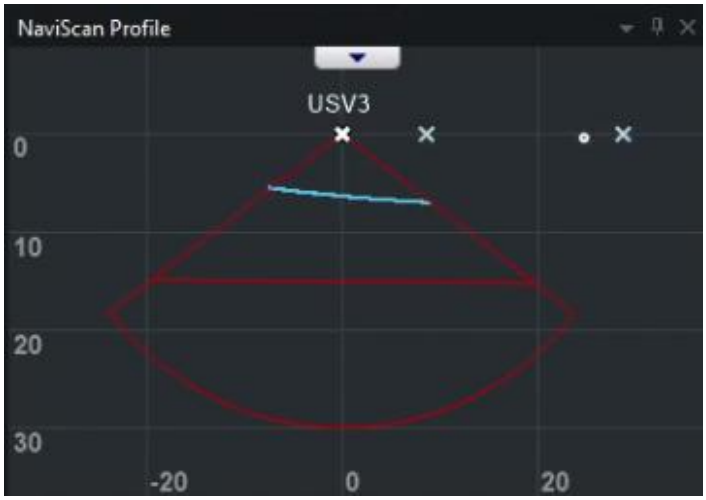


Figure 17 The NaviScan Profile

4 Autopilot interfacing and troubleshooting

When using an autopilot, the autopilot driver has a two-way communication to allow not only for commands to be sent to the autopilot, but also to receive status information. **Port** is for outgoing autopilot messages, while **Extra COM port** is used for receiving autopilot status messages.

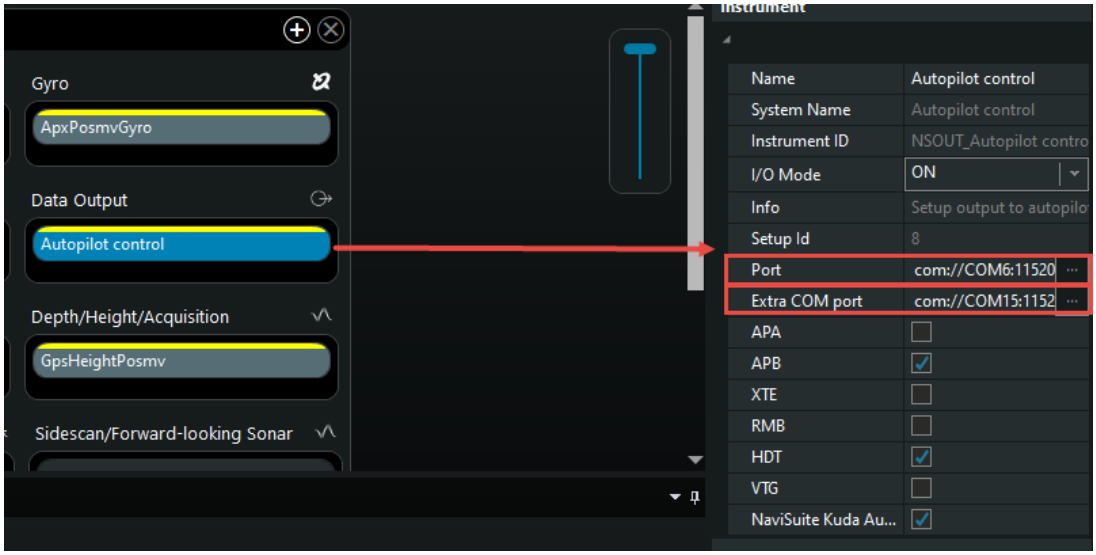


Figure 18 The Autopilot Control driver with the port and Extra COM Port

Note: We recommend sending NMEA APB and HDT as a minimum, with the NMEA VTG being also important for controlling the speed.

The NaviSuite Kuda autopilot messages come from the autopilot. The string is formatted as:

Figure 19 Autopilot string format

The string starts with the *$EIVAAP* string prefix. The numbers in the string are detailed in the screen shot below:

vessel_armed	1
operation_mode	0
manual_steering_rc_pc	1
system_failure_mode_active	0
vessel_vector_forward	0
vessel_vector_sideways	0
vessel_vector_yaw	0
bat_stbd_voltage	32.88
bat_stbd_amp	0.00
bat_stbd_charge	100.00
bat_port_voltage	32.87
bat_port_amp	0.00
bat_port_charge	100.00
bat_general_voltage	0.00
bat_general_amp	-12.50
bat_general_charge	0.00
rc_remote_aux0	0
rc_remote_aux1	0
rc_remote_aux2	0
rc_remote_aux	0
vessel_thruster_working0	1
vessel_thruster_working1	1
vessel_thruster_working2	1
vessel_thruster_working3	1

vessel_heading_present	1
vessel_speed_present	0
runline_present	1
runline_advanced_present	0
vessel_heading	315.63
vessel_speed	0.00
runline_current_dol	9.09
runline_current_heading	23.83
runline_arrival_circle_entered	0
runline_current_distance_to_end	10000.00
runline_next_heading	0.00
runline_next_distance_to_end	10000.00
runline_next_arch_radius	0.00
vessel_target_heading	37.24
usv_target_survey_percentage	30
usv_target_survey_speed	1.30
vessel_thruster0	0
vessel_thruster1	0
vessel_thruster2	0
vessel_thruster3	0

Figure 20 EIVA Autopilot string fields and an example of decoding a string content using DataMon

The following positions are highlighted as some of the most important options to pay attention to:

1. Vessel Armed – 0 (No) or 1 (Yes)
2. Operation Mode – 0 (Idle), 1 (Joystick), 2 (Manual) and 3 (Kuda Core)
3. Manual Steering – 0 (No) and 1 (Yes)
4. System Failure – 0 (no) and 1 (Yes)
10. Battery Starboard Charge – percentage of charge
13. Battery Port Charge – percentage of charge
15. Battery General Charge - percentage of charge

Figure 21 Highlighting the positions of the most important data

The incoming autopilot messages can be seen in NaviSuite Kuda Core. You must enable it in the vehicle **Properties** to send autopilot messages to log.

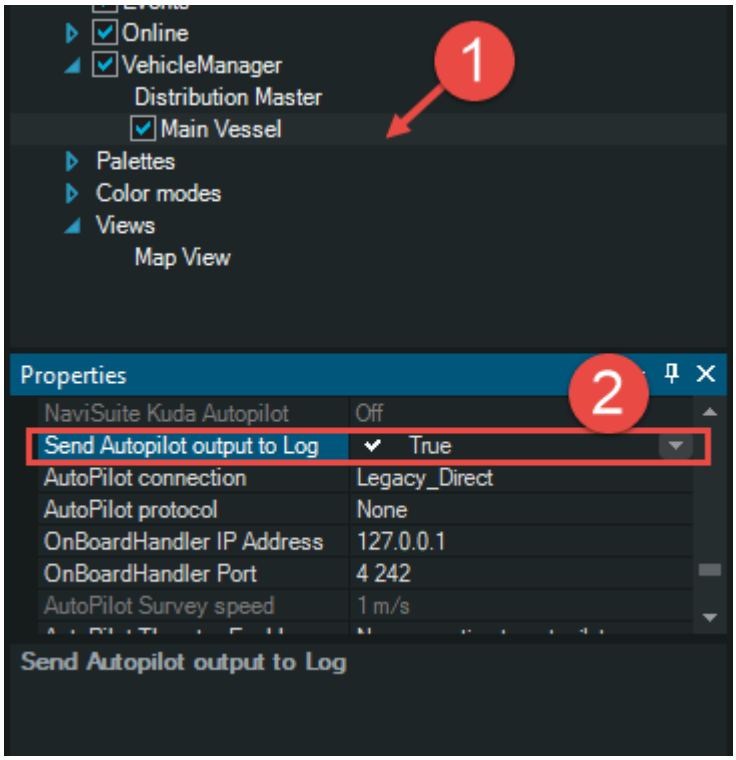


Figure 22 Enabling the sending of autopilot messages in the Log Window

By opening the **Log Window** (either from **View** > **Views** > **Log Window** or Ctrl+L) you can read the actual incoming and outgoing strings from and to the autopilot.

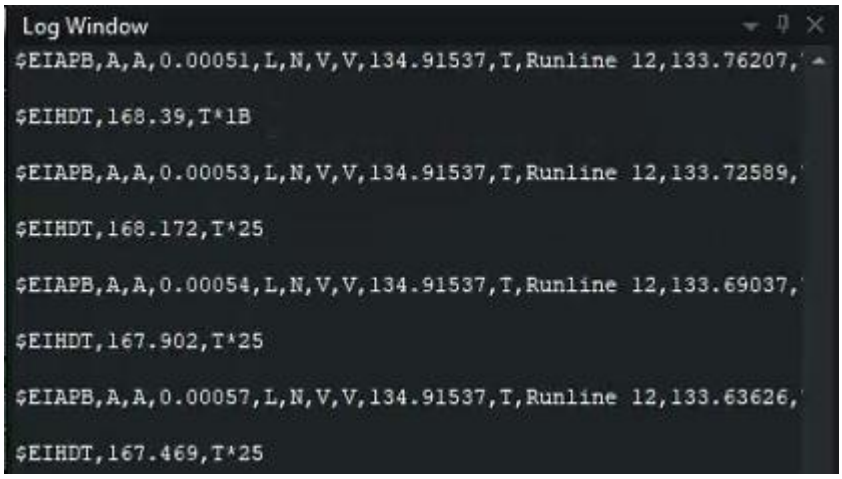


Figure 23 Log window with autopilot messages

5 Example: Step-by-step USV setup and quick start

5.1 Introduction

The example describes the setup and usage of the NaviSuite Kuda Core software to control a USV with a remote computer.

For this situation, you will be taken through the set-up of the onboard USV computer, then a remote computer acting as a supervisor.

NaviSuite Kuda Core will be installed on both computers, however only one licence is required.

The software provides the communication between the computers, allowing the onshore computer to take control of the USV.

5.2 Preparation of the USV computer

5.2.1 Installation

For the best results, EIVA recommends the following:

Tool	Recommendation
Hardware	Windows 11 computer, latest updates
Product	NaviSuite Kuda Core
Software version	KudaCore_4.8
Firewall	Default (the software will add rules, however)

1. Run the installer and use the default path.
    - Two desktop shortcuts are created, **Kuda Core Config** and **Kuda Core**.

Open NaviSuite Kuda Core Config**.**

    - EIVA Licence Activator should also open, otherwise find it in **Help**
2. Activate your NaviSuite Kuda Core licence.
    - See our knowledge base for help getting started.

Note: Add NaviSuite Kuda Core Config to the Windows Startup, to ensure that it is always running .

5.2.2 A simulated setup

If you would like to test the set up before mobilisation, you can follow these steps to make a simulated vehicle.

5. Open NaviSuite Kude Core.
6. Click on the **Main Vessel** and use **+** sign to add instruments.
7. Add the instruments *Free GPS* and *Free Gyro*.
8. Add the *Autopilot control*, set the port to the computer's IP address.
9. Save the project with an identifiable name (eg *USV Setup.NSP*).
10. Start NaviSuite Kuda Core Config.
11. Save the NaviSuite Kuda project and give it a project name.
12. Connect to the local NaviScan, running in the background.
13. Browse to the **View** tab in the **Menu bar** and add **Vehicle Control**.
14. In the **Tool bar**, select **Options** > **Remoting** to enable remoting in NaviSuite Kuda Core Config.
15. Go to the Windows command prompt (Windows + R, write *cmd* + enter). Type *ipconfig* and write down the IP address of the USV computer (you will need it for the onshore computer).
16. In the project title, you can now see that the remote control is enabled.

The result is shown in Figure 24 NaviSuite Kuda Core Config shown with Remote control enabled.

Figure 24 NaviSuite Kuda Core Config shown with Remote control enabled

Check the **Online** node in the **Project Tree**. Green dots represent an active working connection. If the dots are red, remove the **Online** node and connect once more.

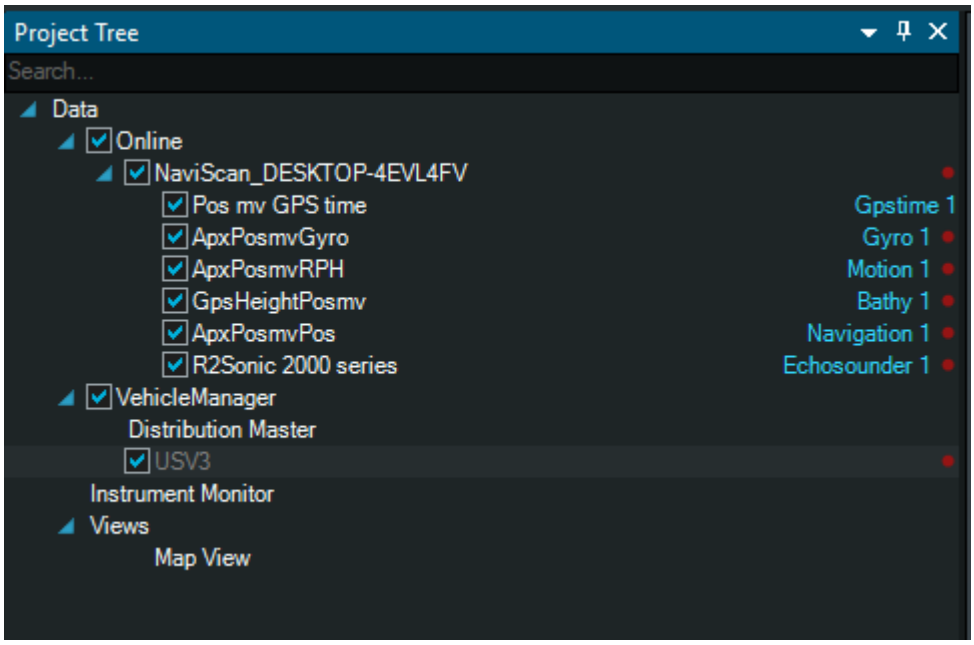


Figure 25 NaviScan not receiving valid data – all instruments are shown with red dots

The **NaviScan logging and settings** toolbar will now be present, with the three options described below.

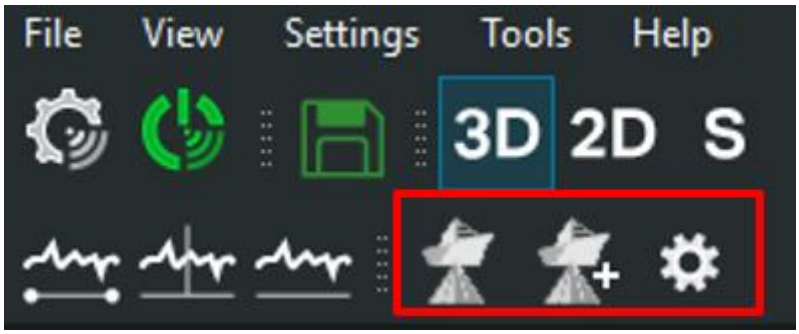


Figure 26 NaviScan logging and settings toolbar

Sbd file logging – Turn on/off logging for the NaviScan SBD file
Split to new sbd file – Starts logging in a new SBD file
Settings – Sonar settings as shown in Figure 15

Your USV computer is now ready to be used. Next, you will set up the supervisor computer.

5.2.3 Example .NSP – USV NaviSuite Kuda Core Config

An example of the NaviSuite Kuda Core Config from the a USV setup tested by EIVA can be found in our knowledge base under Getting started with NaviSuite Kuda Core : EIVA Support. See the NaviSuiteKudaCore.NSP file.

5.3 Preparation of the supervisor computer onshore

For the best results, EIVA recommends the following:

Tool	Recommendation
Hardware	Windows 11 computer with all updates
Licence	No licence needed
Software	The same NaviSuite Kuda Core version that is installed on the USV
Firewall settings	Default

1. Install NaviSuite Kuda Core to the default path.
2. Open NaviSuite Kuda Core Config.
3. Enable **Remoting**.
4. Use the IP address and port from the USV computer.
5. Set to **Supervisor**.
6. Download the configuration from the USV.
7. Open Kuda Core Config and set the mode to **Supervisor**.

Now the supervisor can create runlines and upload them to the USV. Open the **Vehicle Control** to load the runline and start your operation.

You can only start the supervisor computer if a master is already running and you can connect to that computer via the external server.

Figure 27 The Connect and Remoting options in the tool bar

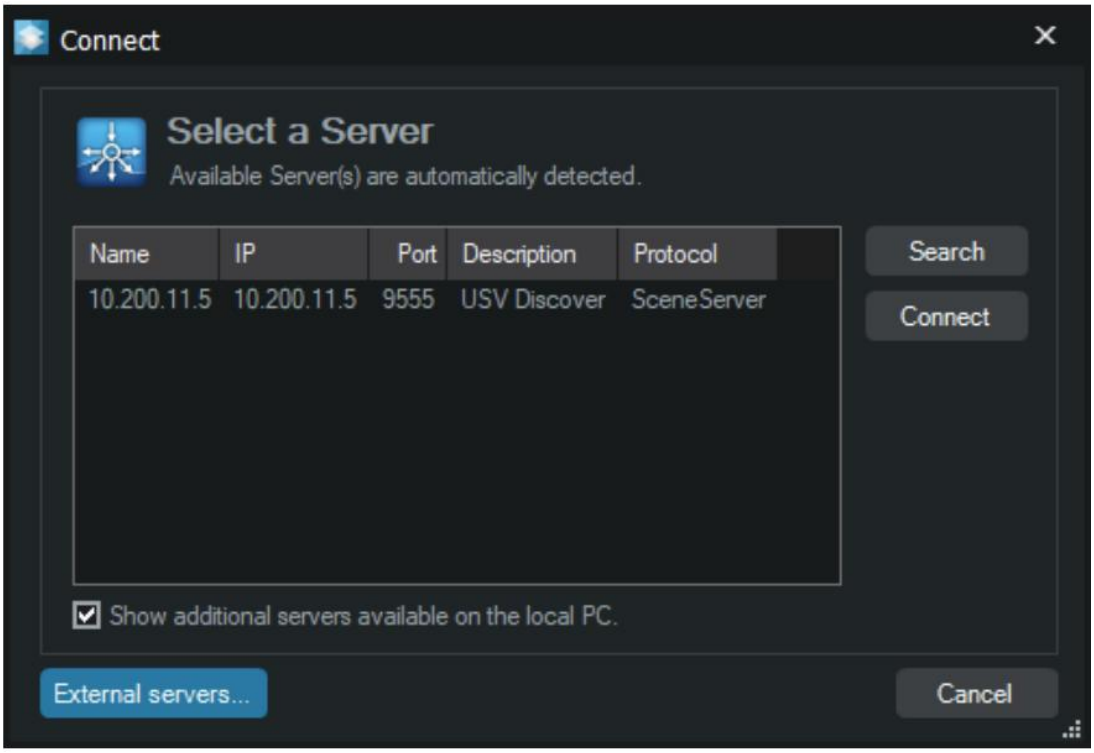


Figure 28 Select a Server

Add the USV IP address, to the external server list. Below is an example of adding 10.200.11.5:9555; SceneServer; USV Discover to the external servers list.

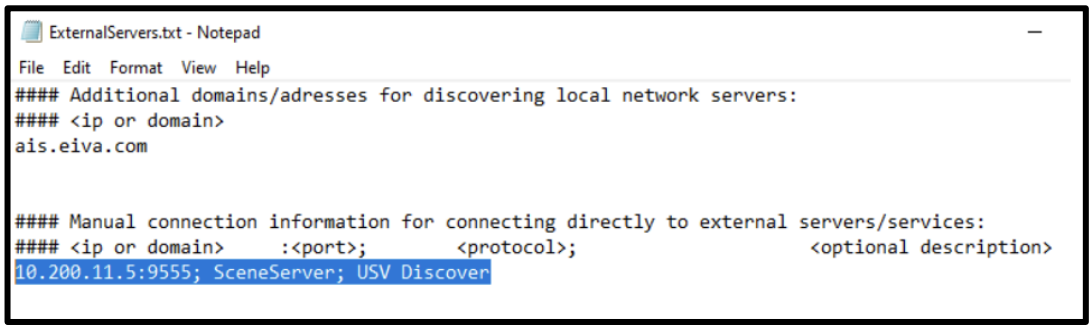


Figure 29 Adding to external servers

5.4 Firewall

NaviSuite Kuda Core will make some changes to your firewall rules. Below are the settings that will be automatically set, and the ports that need to be manually opened.

5.4.1 Automatically applied rules

Inbound Rules

Name	G...	Profile	Enabled	Action	Override	Progr
EIVA Remote Service Open TCP		All	Yes	Allow	No	C:\EI\
EIVA Remote Service Open UDP		All	Yes	Allow	No	C:\EI\
EivaMaster Open TCP		All	Yes	Allow	No	C:\EI\
EivaMaster Open UDP		All	Yes	Allow	No	C:\EI\
Kuda Core Open TCP		All	Yes	Allow	No	C:\EI\
Kuda Core Open UDP		All	Yes	Allow	No	C:\EI\
NaviScan Config Open TCP		All	Yes	Allow	No	C:\EI\
NaviScan Config Open UDP		All	Yes	Allow	No	C:\EI\
NaviScan Open TCP		All	Yes	Allow	No	C:\EI\
NaviScan Open UDP		All	Yes	Allow	No	C:\EI\
NaviSuite Master Remote Services		All	Yes	Allow	No	Any
NaviSuiteConfig Open TCP		All	Yes	Allow	No	C:\EI\

Figure 30 Firewall settings set by NaviSuite Kuda Core

5.4.2 Manually opened ports on 4G router

Be sure to open the ports listed in the getting started guide for NaviSuite Kuda Core.

Port Mapping

Enable	Proto	Source	Service Port	Internal Address	Internal Port	Log	External Interface(Optional)	External Address(Optional)	Description
Yes	TCP & UDP	80.160.68.90	9555	192.168.2.6	9555	No	Cellular 1		cellulart usv
Yes	TCP	80.160.68.90	6547	192.168.2.6	6547	No	Cellular 1		Kuda config supervision
Yes	TCP	80.160.68.90	2000	192.168.2.6	2000	No	Cellular 1		NS
Yes	TCP	80.160.68.90	31231	192.168.2.6	31231	No	Cellular 1		HMD services (EivaMaster.exe)
Yes	UDP	80.160.68.90	5999	192.168.2.6	5999	No	Cellular 1		NaviScan center depth
Yes	UDP	80.160.68.90	2300	192.168.2.6	2300	No	Cellular 1		NaviScan DTM (reduced MBE)
Yes	TCP & UDP	80.160.68.90	2301	192.168.2.6	2301	No	Cellular 1		NaviScan NED exporter
Yes	TCP & UDP	80.160.68.90	2302	192.168.2.6	2302	No	Cellular 1		NS SS DTM
Yes	TCP & UDP	80.160.68.90	2304	192.168.2.6	2304	No	Cellular 1		NS NM scan writer
Yes	TCP	80.160.68.90	2305	192.168.2.6	2305	No	Cellular 1		NS HMD
Yes	TCP	80.160.68.90	2100	192.168.2.6	2100	No	Cellular 1		Real-time DTM (from NS to ?)
Yes	TCP & UDP	80.160.68.90	9777	192.168.2.6	9777	No	Cellular 1		NS dynamic port 1
Yes	TCP & UDP	80.160.68.90	9778	192.168.2.6	9778	No	Cellular 1		NS dynamic port 2
Yes	TCP & UDP	80.160.68.90	554	192.168.2.248	554	No	Cellular 1		IP-cam
☑	TCP	0.0.0.0/0	8080		8080	☐			

Figure 31 Manually opened 4G router connections

6 Example: Step-by-step manned vessel setup and quick start

6.1 Introduction

The example describes the setup and usage of the NaviSuite Kuda Core software for a manned vessel. For this situation, you will be taken through the set-up of the onboard computer.

6.2 Preparation of the onboard computer

6.2.1 Installation

For the best results, EIVA recommends the following:

Tool	Recommendation
Hardware	Windows 11 computer, latest updates
Product	NaviSuite Kuda Core
Software version	KudaCore_4.8
Firewall	Default (the software will add rules, however)

1. Run the installer and use the default path.
    - Two desktop shortcuts are created, **Kuda Core Config** and **Kuda Core**.
2. Open NaviSuite Kuda Core Config**.**
    - EIVA Licence Activator should also open, otherwise find it in **Help**
3. Activate your NaviSuite Kuda Core licence.
    - See our knowledge base for help getting started.

Note: Add NaviSuite Kuda Core Config to the Windows Startup to ensure that it is always running .

6.2.2 A simulated setup

If you would like to test the set up before mobilisation, you can follow these steps to make a simulated vehicle.

1. Open NaviSuite Kude Core.
2. Click on the **Main Vessel** and use **+** sign to add instruments.

3. Add the instruments *Free GPS* and *Free Gyro*.
4. Add the *Autopilot control*, set the port to the computer's IP address.
5. Save the project with an identifiable name (eg *VesselSetup.NSP*).
6. Start NaviSuite Kuda Core Config.
7. Save the NaviSuite Kuda project and give it a project name.
8. Connect to the local NaviScan, running in the background.
9. Browse to the **View** tab in the **Menu bar** and add **Vehicle Control**.

The NaviSuite Kuda Core Config view should look like this:

Figure 32 NaviSuite Kuda Core Config shown for a manned vessel

Check the **Online** node in the **Project Tree**. Green dots represent an active working connection. If the dots are red, remove the **Online** node and connect once more.

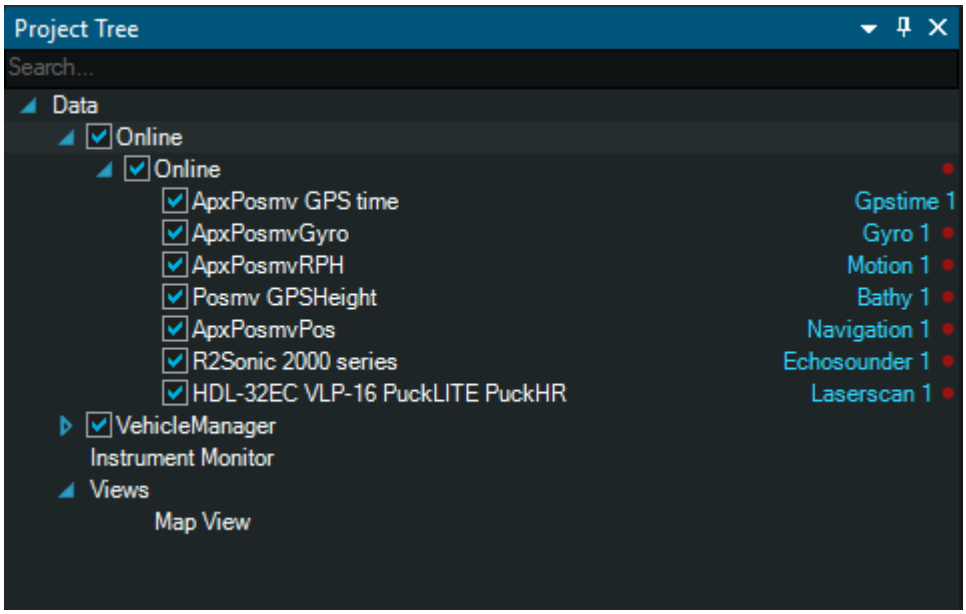


Figure 33 NaviScan is not receiving valid data – all instruments are shown with red dots

The **NaviScan logging and settings** toolbar will now be present, with the three options described below.

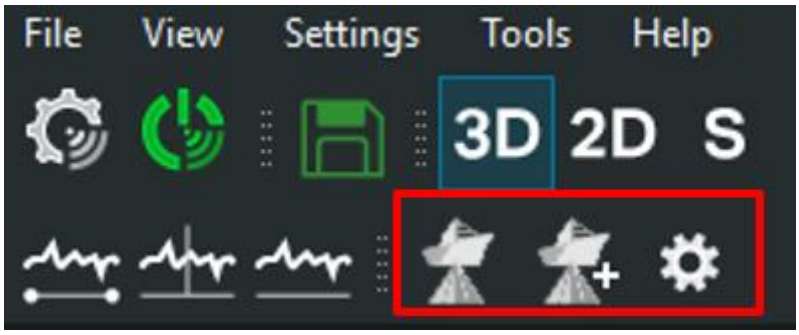


Figure 34 NaviScan logging and settings toolbar

Sbd file logging – Turn on/off logging for the NaviScan SBD file
Split to new sbd file – Starts logging in a new SBD file
Settings – Sonar settings as shown in Figure 15

Your NaviSuite Kuda Core computer is now ready to be used.

7 Troubleshooting and FAQs

Q. Where are crash logs stored?

A. Crash logs are stored in the folder *C:\ProgramData\EIVA\Kuda Core\Diagnostics*

Q. NaviScan is running as a hidden process by default. Can I change that?

A. Yes. Enabling the tick box **Show Naviscan** will start NaviScan as a normal process, not as hidden process. This option can be enabled in NaviSuite Kuda Core Config > **Project Settings** > **Online** > **Advanced**.

Q. What can I do when NaviScan shows an alarm about lacking privileges?

A. The user has no administrator rights on the computer. Try to set the Windows option *Run as administrator* for NaviScan.

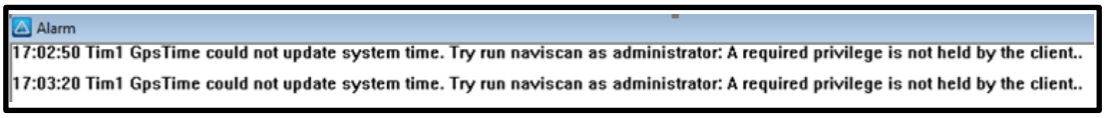


Figure 35 NaviScan running as a user

Q. The supervisor freezes temporarily when uploading waypoints or runlines. How can I handle this?

A. Try to delete obsolete data from the USV and supervisor computer.

1. On the USV, remove the files from the project via the **Project** tree.
2. Save the project.
3. Go to the **Project** tree > right click **Open containing folder** > delete the files there.
4. Switch to the Supervisor computer. Repeat step three.

Q. Which services should be running?

A. The following services should be running:

EIVA Master Service
EIVA Remote Service
EIVA Time Service